Marine Microbial Biodiversity

Lecture 11

Microbial Ecology

Matteo Selci, PhD

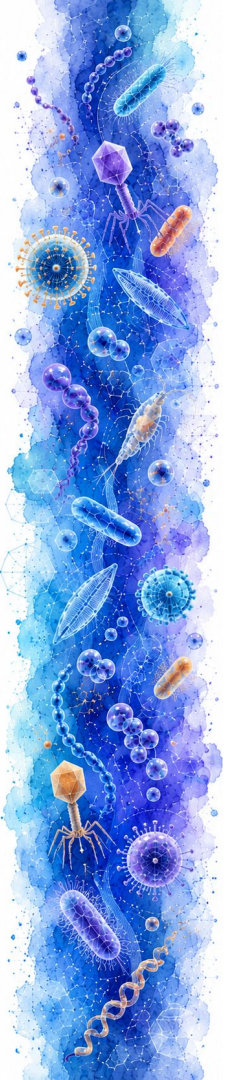
matteo.selci@unina.it

www.donatogiovannelli.com

@matteo.selci

@matteo_selci

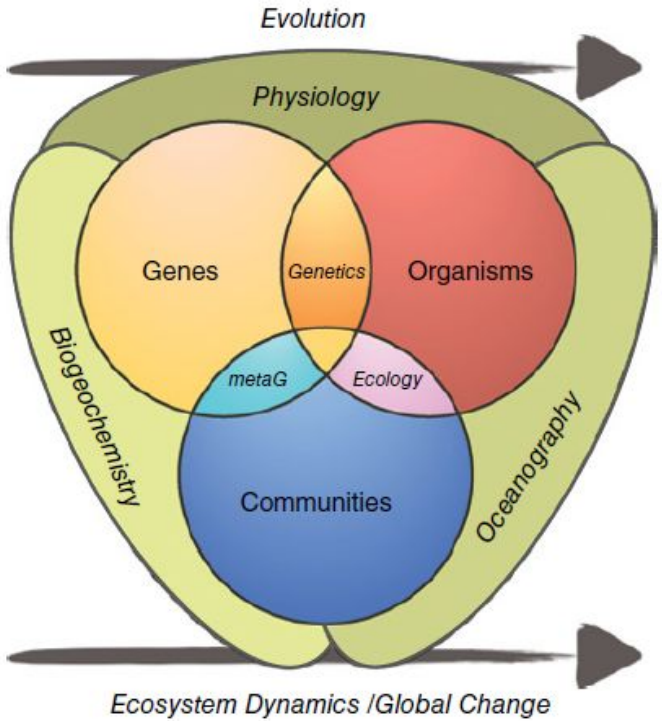




Microbial Lifestyles

Microbial ecology is the **study of the interactions of microorganisms with their environment**, each other, and plant and animal species. It includes the study of symbioses, biogeochemical cycles and the interaction of microbes with anthropogenic effects such as pollution and climate change.

Marine microbial ecology is one of the most dynamic scientific fields because it is at the crossroads of many disciplines such as oceanography, biogeochemistry, microbiology (including protistology and virology), physiology, evolution, and genomics.

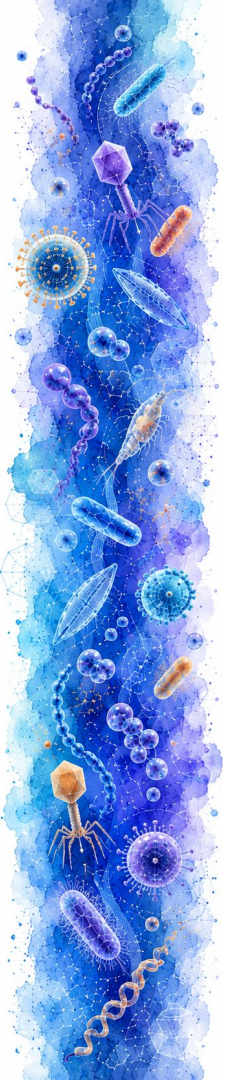


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Ecosystems



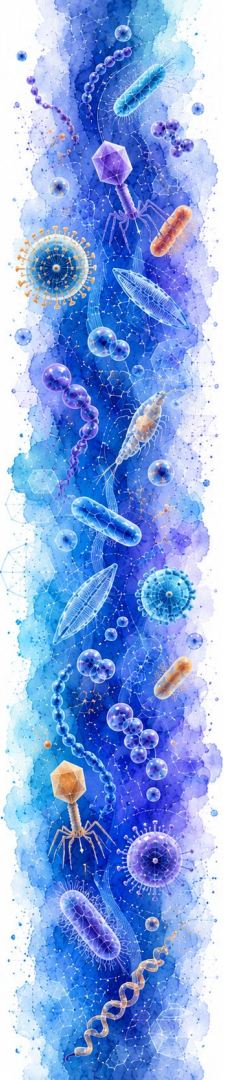
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What is an ecosystem?

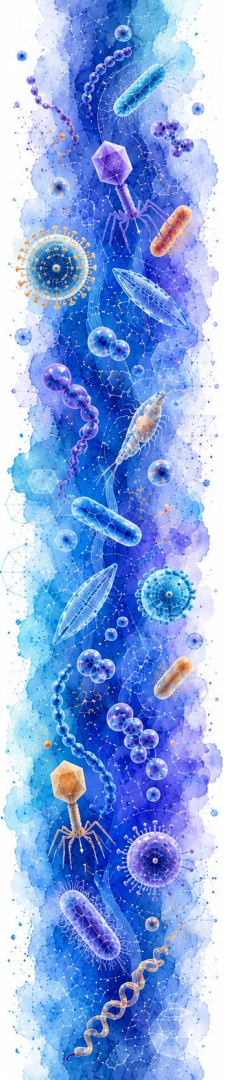


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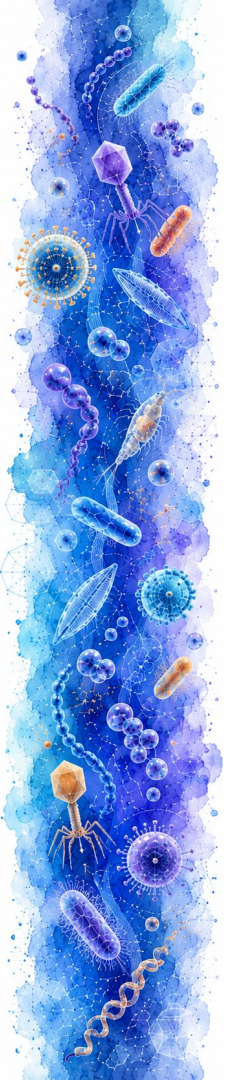


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Ecosystems

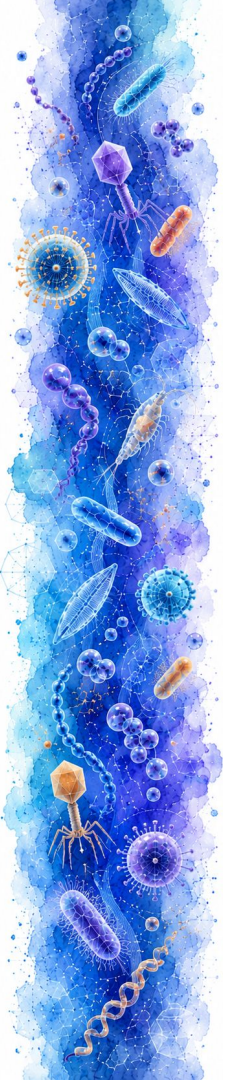
The term "ecosystem" is a composite name: from greek oikos (**οἶκος**, family, family property, house) and the english *system*. Oikos was the *basic unit* of greek society.

In ecology, refers to **a biological community of interacting organisms and their physical environment, in a complex network, and interconnected system**.

First used in 1935 by *Sir Arthur George Tansley*, English botanist and pioneer in the science of ecology, to describe the transfers of materials between organisms and their environment.
He later refined the term, describing it as "The whole system, ... including not only the organism-complex, but also the whole complex of physical factors forming what we call the environment".

In an ecosystem, organisms and their environments can also be described as **biotic and abiotic components linked together through nutrient cycles and energy flows**.

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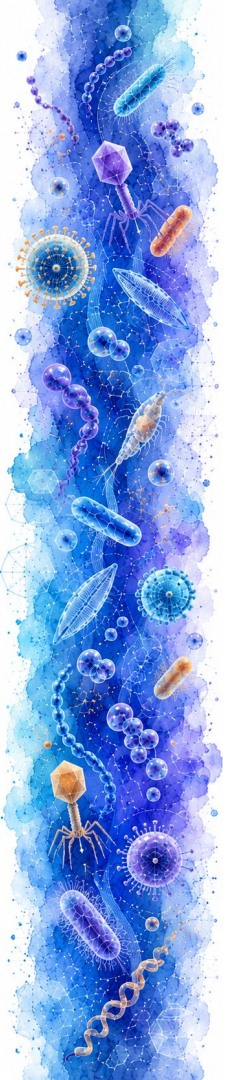
Ecosystems

Collectively, microorganisms show great metabolic diversity and are the primary catalysts of nutrient cycles in nature. The **types of microbial activities** possible in an ecosystem are a function of the **species present**, their **population sizes**, and the **physiological state** of the microorganisms in each habitat. By contrast, the **rates of microbial activities** in an ecosystem are controlled by the **nutrients** and **growth conditions** that prevail.

A **group of microorganisms** of the **same species** that reside in the **same place** at the **same time** constitutes a **microbial population** and may be descendants of a single cell.
A **community** consists of **populations of one species** living in **association** with **populations of one or more** other species. The microbial species that reside in a certain habitat are those best able to grow with the nutrients and conditions that prevail there.

The diversity of microbial species in a community can be expressed in two ways. One is **species richness**, the total number of different species present. **Species abundance**, by contrast, is the proportion of each species in the community. Species richness may also be expressed in molecular terms by the diversity/proportion of phylotypes (16S rRNA) observed in a given community.

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Species Concept

In biology, a species is often defined as the largest group of organisms in which any two individuals of the appropriate sexes or mating types can produce fertile offspring, typically by sexual reproduction. This species concept works well for sexually reproducing organisms, but may be useless for species that reproduce asexually, such as bacteria.

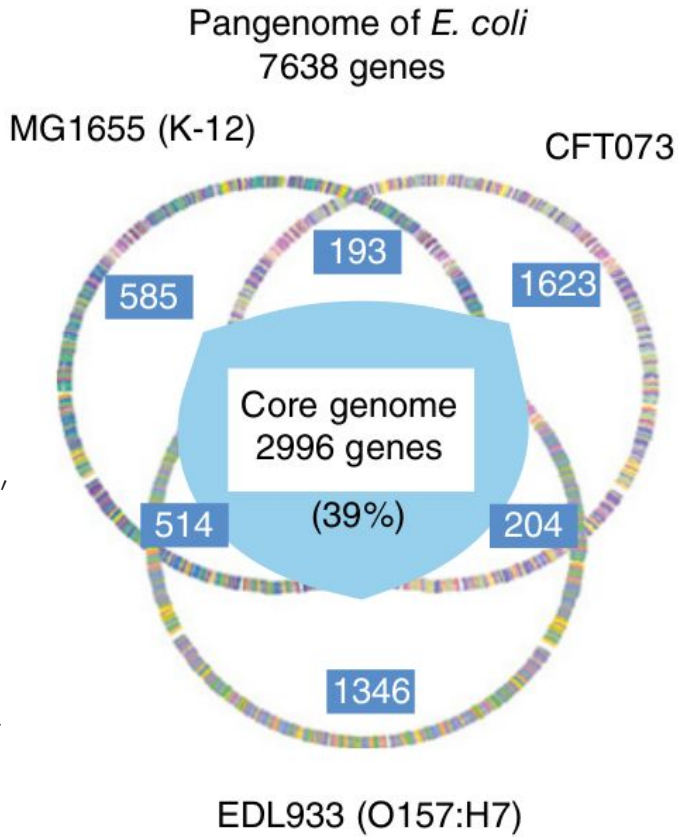
The current standard for **defining a species** is a **polyphasic approach**, combining **phylogenetic**, **physiology** and **chemotaxonomy**, which focus on the analyses of two similarities: (1) **16S rRNA gene**, since it is a conserved universal target not subject to horizontal gene transfer, or (2) whole genome similarity, also called ANI, **Average Nucleotide Identity**. Current standard boundaries for the same species are: 16S rRNA similarity **> 97%** and whole genome ANI **> 95%** **(=digital DNA-DNA hybridization (dDDH) > 70%)**.

However, both the approaches described to delineate prokaryotic species are dependent on arbitrary thresholds, and their current use is limited to genomes from cultured isolated microorganisms.
It is well known that many of the cultured marine microbes are not really abundant in the ocean, and moreover, many of the ecologically relevant taxa and metabolic key players are still uncultured and unknown.

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Species Concept

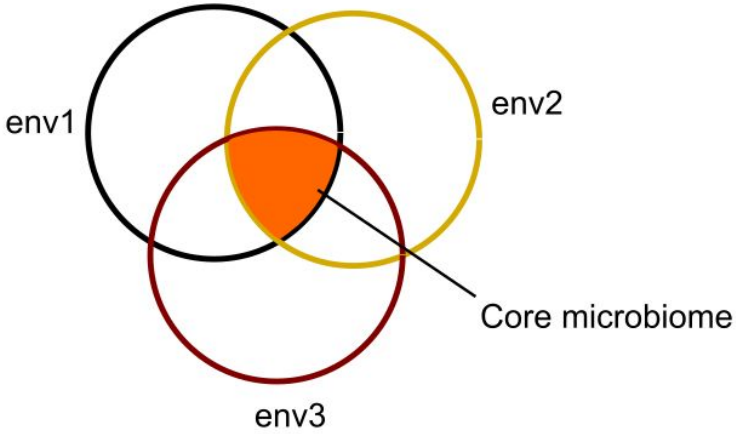
In the fields of molecular biology and genetics, a "**Pangenome**" is the entire set of genes from all strains within a clade. It is represented with a Venn diagram, generally used to show the logical relation between sets. In a pangenome, the **Core genome** represents the set of homologous genes that are present in all genomes of an analyzed dataset. In defining a species, the core genome are those genes shared by all the strains of that species.
Not always SPECIES AFFILIATION = HIGH SIMILARITY
The *Escherichia coli* pangenome includes genes from K-12, CFT073 (uropathogenic) and EDL933 (enterohemorrhagic) strains. Notwithstanding they are the same species, these three strains share only 39% of their genome. Hence, for the standard role in microbial taxonomy they should be separated species, instead they are the species because at a physiological level, at a metabolic level and at a chemotaxonomic level they are very similar.



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Community Structure: Core Microbiome

$$\text{Core} = A \cap B \cap C$$
$$\text{Shell} = A \cap B + A \cap C + B \cap C$$
$$\text{Cloud} = (A \cup B + A \cup C + B \cup C) - (A \cap B \cap C)$$
$$\text{Pan} = A \cup B \cup C$$

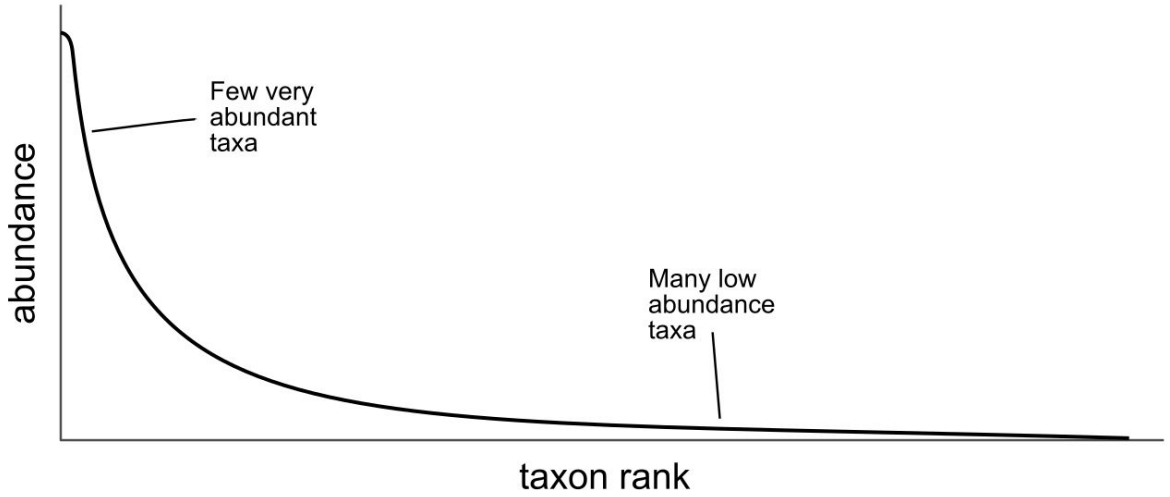


The **core** microbiome represents **the taxa present in all the observed samples**. It can be defined across an environment (core microbiome in surface sediments), for a selected area (the core microbiome of the Antarctic waters) or for a specific functions (the core microbiome carrying out nitrogen fixation). A similar concept can be used for functional diversity (the core functions instead of taxa). It can be seen as the deterministic portion of the community. The complementary portion of the core is composed by the **shell** and the **cloud** microbiome, whereas the union of all the taxa across the samples is the **panmicrobiome**.

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Community Structure: Microbial Seed-Banks Theory

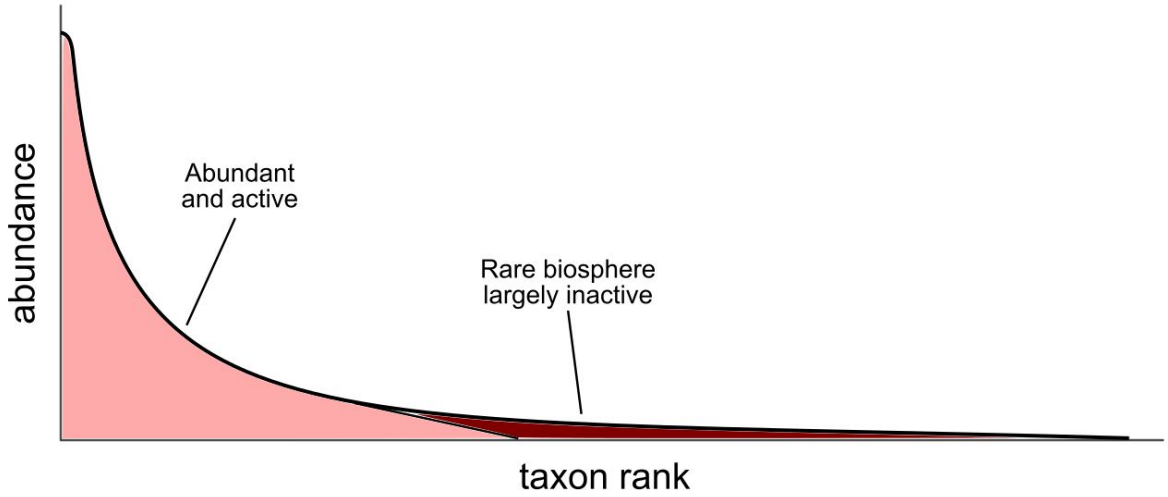
The rank-abundance curve or **Whittaker plot** is a chart used to display relative species abundance, species richness and species evenness of a given community in a given environment. The resulting graph is a curve with a long-tail distribution: **very few species are abundant, while a lot of species have very low abundance**.



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Community Structure: Microbial Seed-Banks Theory

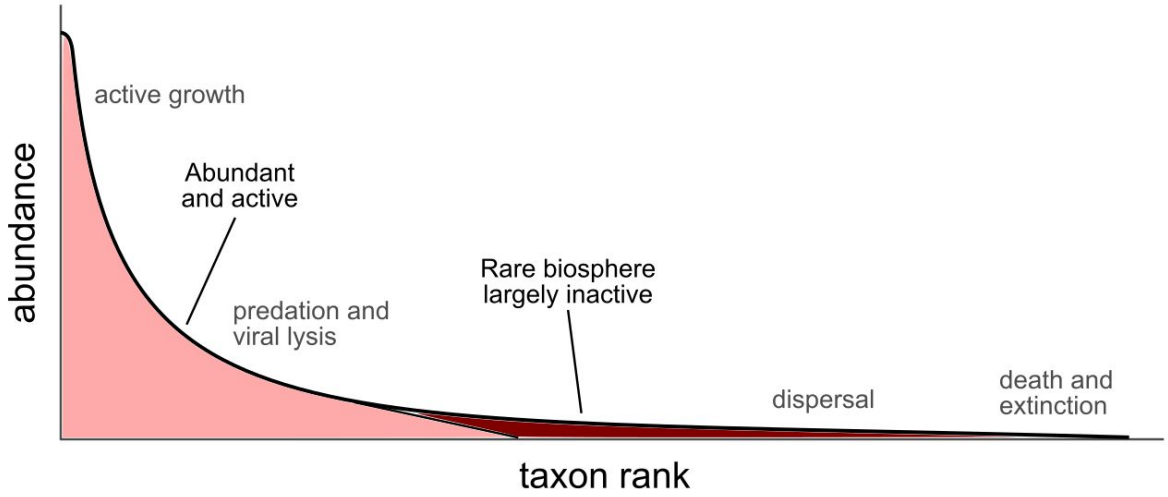
Microbial seed-banks theory: Relatively few microbial taxa are abundant and active, while the large majority of the diversity is constituted by a large number of low abundance taxa, often in a dormancy state. Dormancy is a bet-hedging strategy. Dormant microorganisms generate a seed bank capable of being resuscitated following environmental change. In other words, rare biosphere is a bank of biodiversity waiting for better times.



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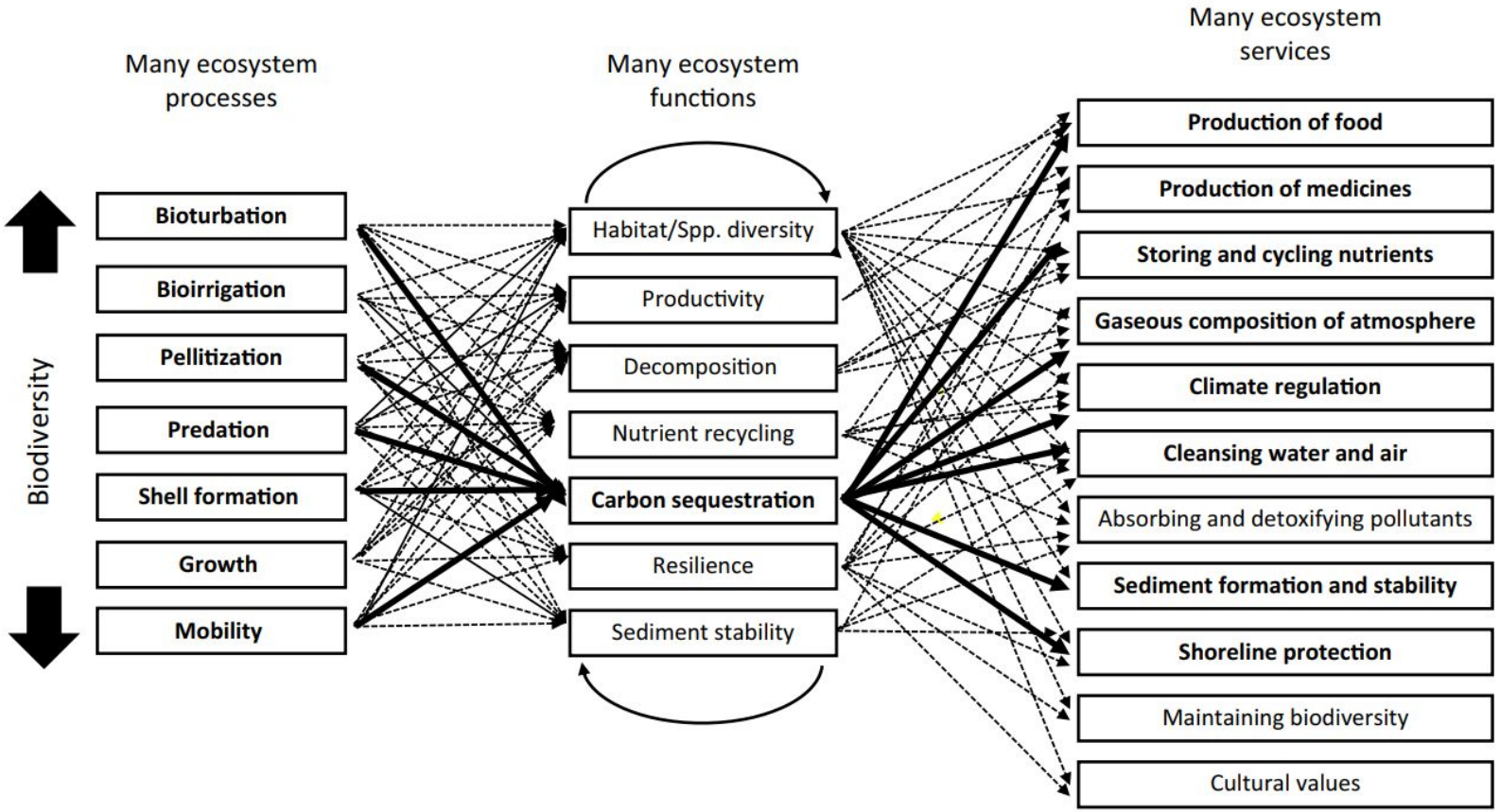
Community Structure: Microbial Seed-Banks Theory

The ecological, biological or geochemical processes acting on the microbial community vary depending on the abundance of the taxa, and therefore on their physiological and metabolic contribution to the ecosystem. For example, the most abundant and metabolically active pool of the community is most affected by **predation and viral lysis**, whereas the rare biosphere undergoes **dispersion** phenomena such as lateral transfer, or **death** and **extinction**.



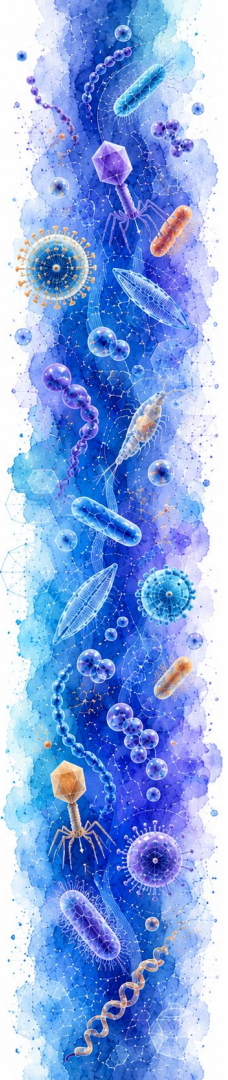
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Biodiversity-Ecosystem Functioning



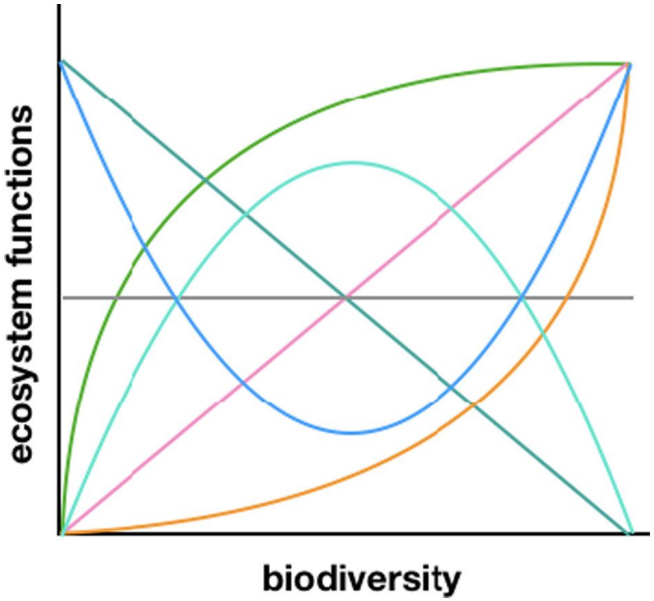
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10.1016/j.tree.2014.05.002



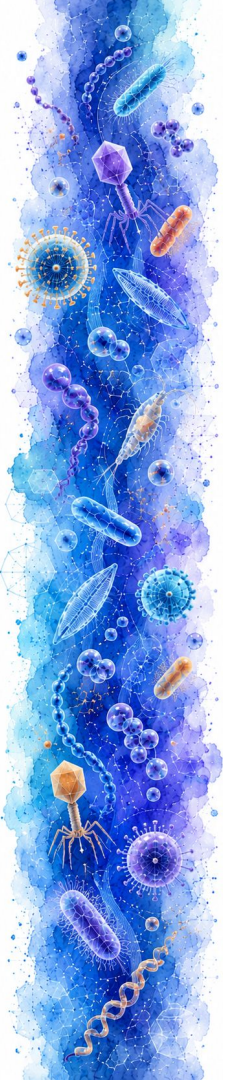
Biodiversity-Ecosystem Functioning

"Ecosystem function" is the technical term used to define the **biological**, **geochemical** and **physical** processes and components that take place or occur within an ecosystem, that control the fluxes of energy, nutrients and organic matter through an environment. They relate to the structural components of an ecosystem (e.g. vegetation, water, sediment, atmosphere and biota) and how they interact with each other, within ecosystems and across ecosystems. Given the key role of microorganisms in element and nutrient cycling, the ecosystem functions provided by the **microbial community are fundamental** for the functioning of the ecosystem, and vice versa.



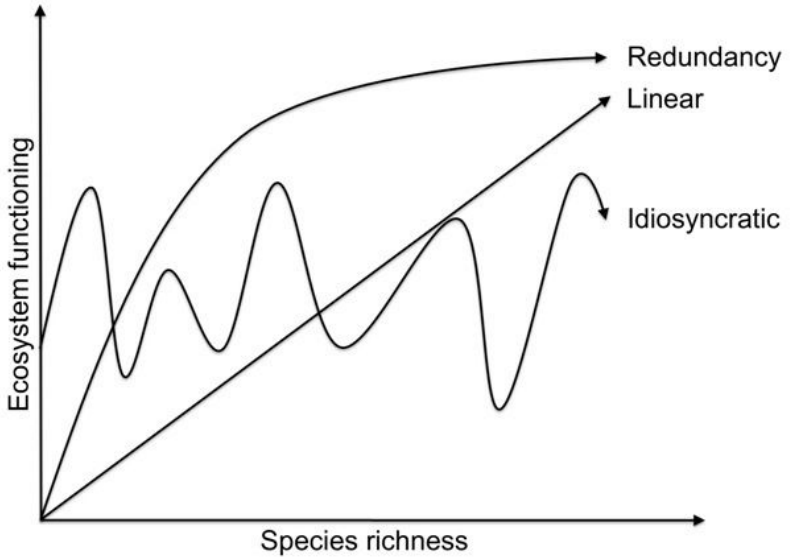
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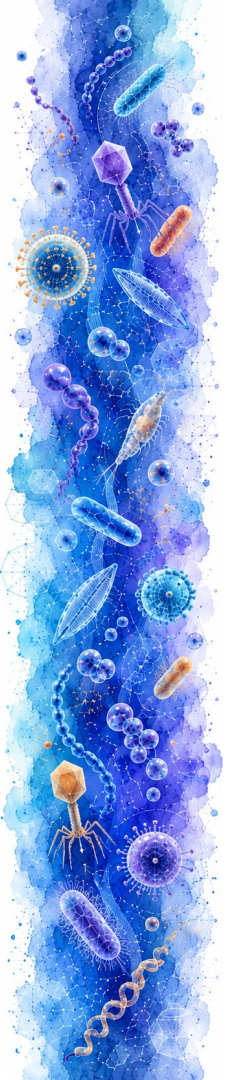


Biodiversity-Ecosystem Functioning

The relationship between biodiversity and ecosystem functioning (called **BEF relations**) are widely studied to understand the role of different taxa and the overall functioning and vulnerability of an ecosystem. Decades of scientific debate have attempted to conceptualize, experientially determine, and model the role biodiversity plays in determining ecosystem functioning. Scientific community has attempted to reconcile **theoretical and experimental findings**, by determining the effects of biodiversity in various ecosystems and the mechanisms underlying those effects, and refining the spatial and biological scales at which diversity is measured.



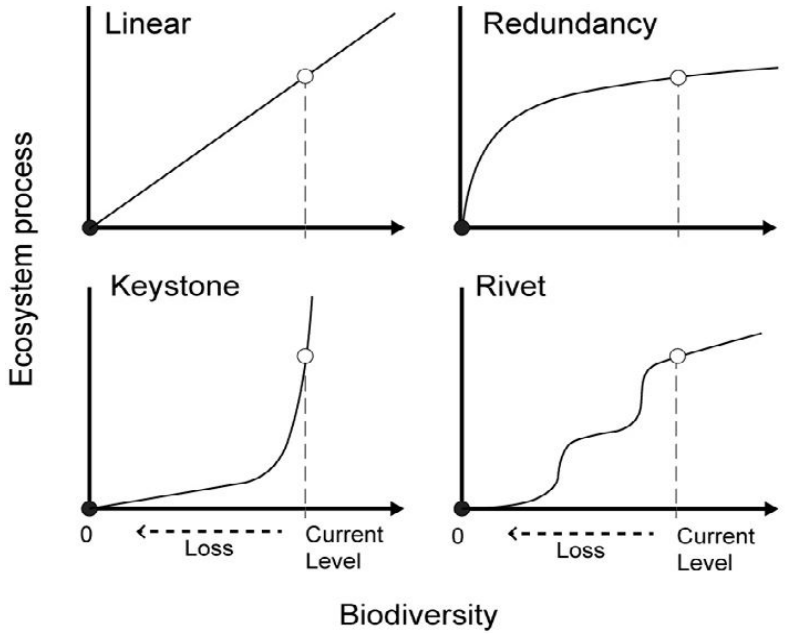
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Biodiversity-Ecosystem Functioning

Early hypotheses suggested linear, redundant, keystone, or rivet response models to explain the relationship between a biodiversity gradient and a given ecosystem process. Although the field has moved beyond these simplified models, they are still archetypes referenced during analyses of contemporary findings.

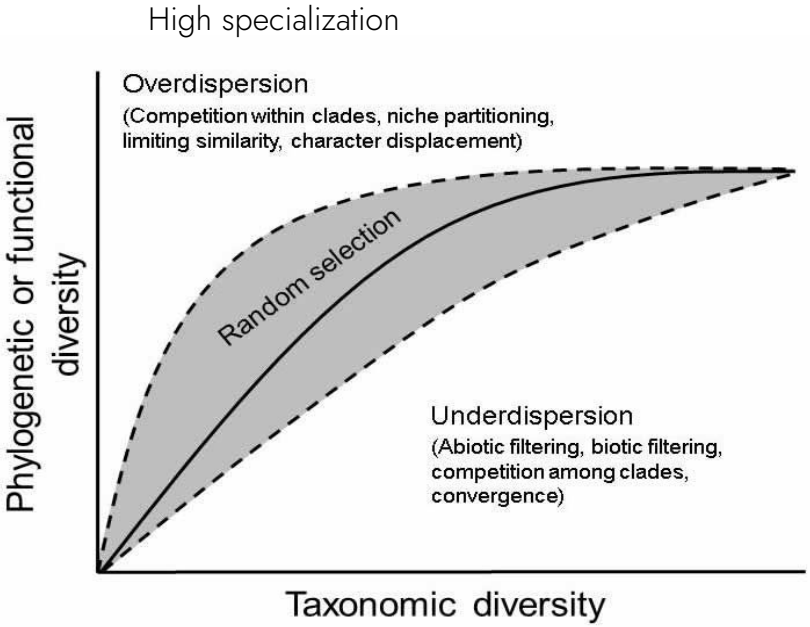
A **linear** response indicates that there is a direct relationship between the biodiversity and the process of interest. **Redundant** models assume species have similar effects on ecosystem processes, causing an asymptotic function. **Keystone** and **rivet** response models (idiosyncratic) suggest a tipping point where if a critical species is lost (keystone) or if a diversity threshold is crossed (rivet), there will be a sharp decline or a drastical ecosystem change.



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BEF: Functional Redundancy

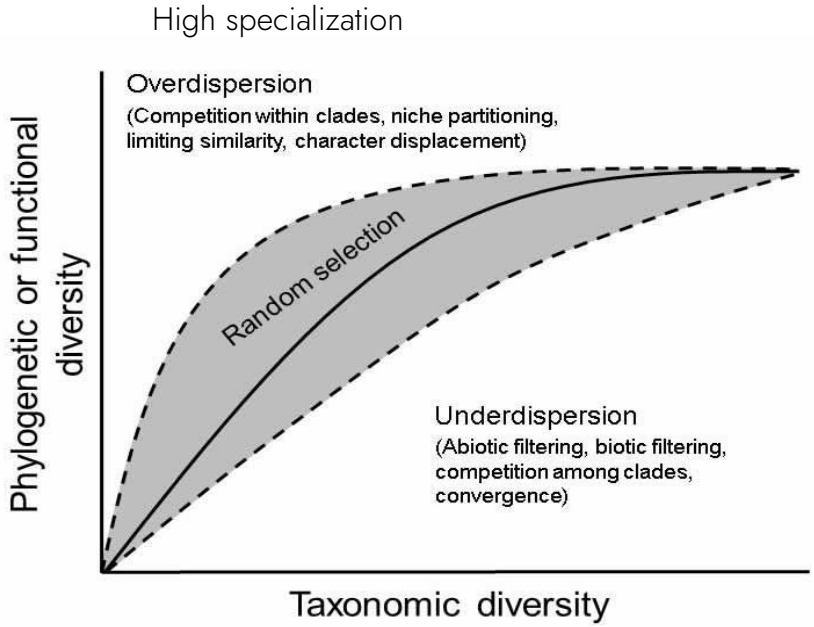
In ecology, **functional redundancy** is the phenomenon describing how multiple species representing a variety of taxonomic groups can share similar, if not identical, roles in ecosystem functionality. Simultaneous consideration of abundance, evolutionary history, and ecological functions of species can define the patterns of biodiversity in space and time.
Deviations from the random selection of species provide evidence for the relative importance of ecological and evolutionary mechanisms that contribute to the structure of a given community. In other words, depending on how a given microbial community distribute in terms of functional biodiversity, we can understand which phenomena are shaping the community.



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BEF: Functional Redundancy

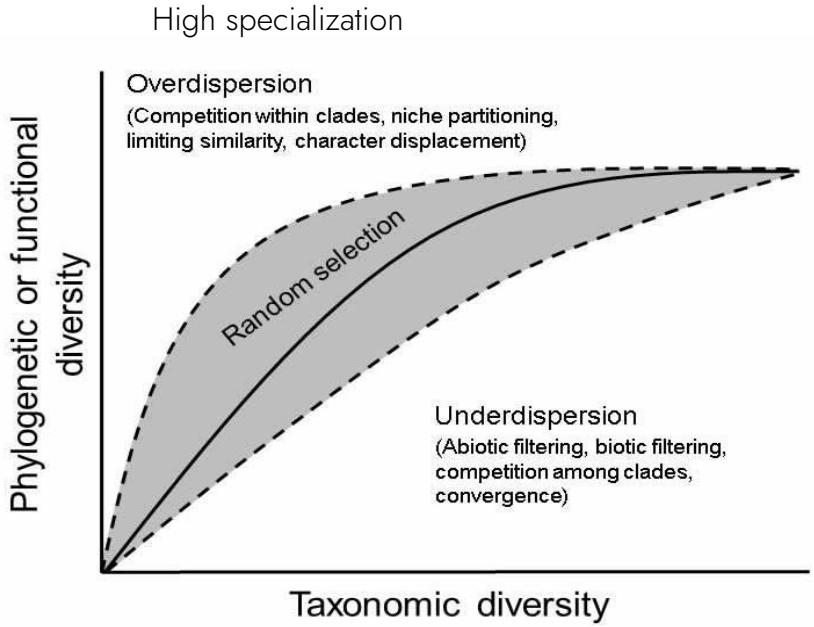
Phylogenetic overdispersion (more phylogenetic diversity than expected) is consistent with community assembly dominated by competitive exclusion of closely related species.
In contrast, **phylogenetic underdispersion** (less phylogenetic diversity than expected) is consistent with community assembly driven by environmental filtering or with interclade competition. As an example, an entire clade may have an advantage over other clades because of superior competitive abilities or because of phylogenetically-conserved adaptations to local environmental conditions.



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BEF: Functional Redundancy

Functional overdispersion (more functional diversity than expected) is consistent with community assembly driven by niche partitioning, limiting functional similarity, or character displacement.
In contrast, **functional underdispersion** (less functional diversity than expected) is consistent with environmental filtering. Environmental filters related to physiological tolerances, habitat affinities, or resource requirements may dominate community assembly, resulting in a less functionally diverse group of coexisting species than would be expected by chance.



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BEF: Ecosystem Disturbance And Recovery

In ecology, a **disturbance** is a temporary change in environmental conditions that causes a pronounced change in an ecosystem.
Disturbances often act quickly and with great effect, to alter the physical structure or arrangement of biotic and abiotic elements.
A disturbance can also occur over a long period of time and can impact the biodiversity within an ecosystem.
Major marine ecological disturbances may include flooding (river input), storms (wave action), outbreaks (pathogens) and trampling (physical destruction).
Earthquakes, various types of volcanic eruptions, tsunami, firestorms, impact events, climate change, and the devastating effects of human impact on the environment (anthropogenic disturbances) such as clearcutting, forest clearing and the introduction of invasive species can be considered major disturbances.





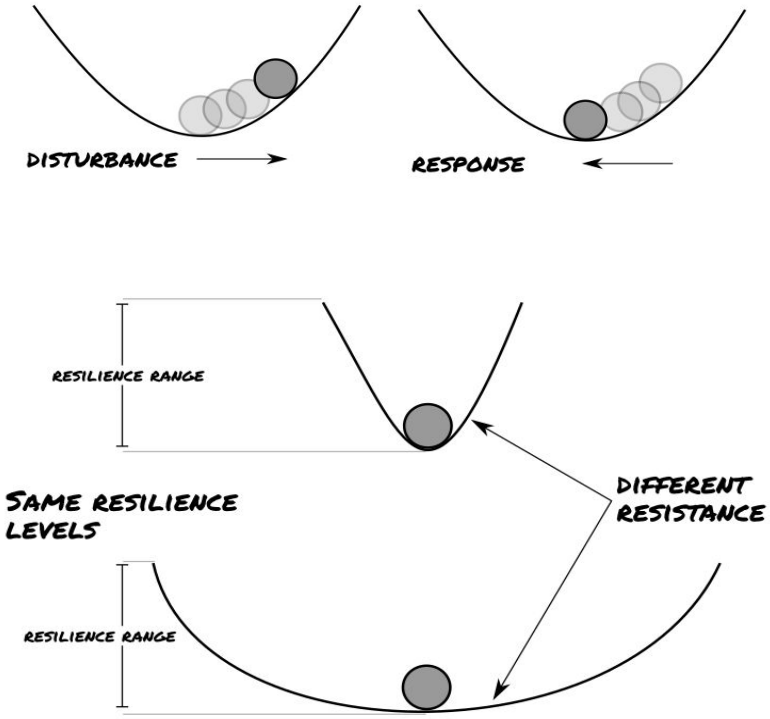
Coral bleaching as marine ecosystem disturbance example

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BEF: Ecosystem Disturbance And Recovery

On the basis of the response of an ecosystem during and after a disturbance it is possible to define the **ecosystem resistance** as the ability to persist or withstand a disturbance, and **resilience** as the ability to recover once a disturbance ends. The extent of a disturbance and the ability of an ecosystem to resist and/or resile to it can be described by a curve: the **slope** indicates the resistance capacity and the **amplitude** the resilience capacity.

Example: We have a ball moving in a valley, the center of which represents a stable state. The disturbance is a factor or an event able to push away the ball. The resistance (slope) is the force required to move the ball, while the resilience is the capacity come back to the center of the valley (stable state) and maintain the ball in it.

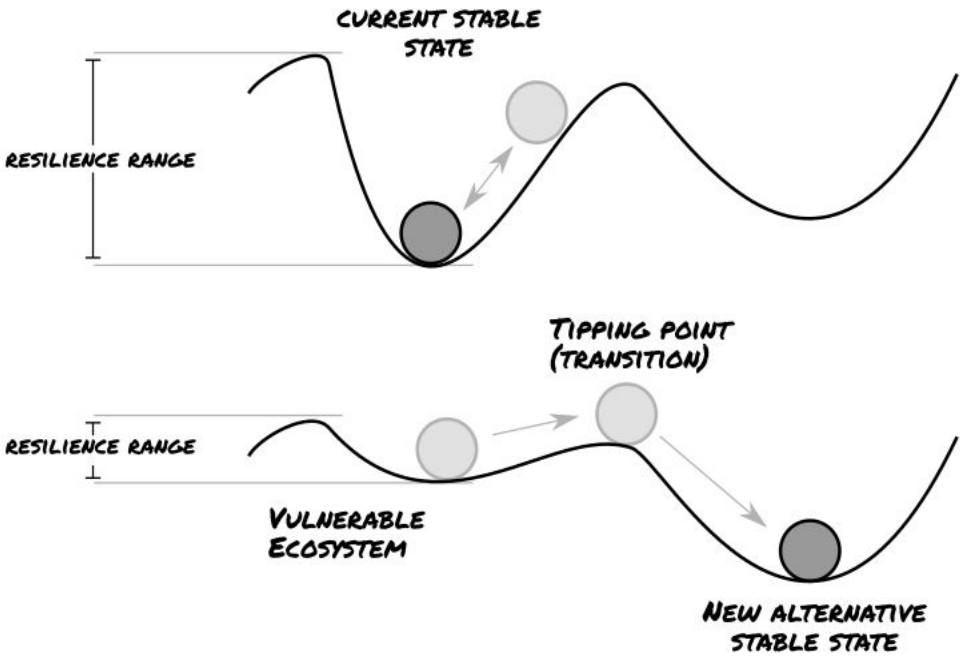


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BEF: Ecosystem Disturbance And Recovery

Most ecosystem state are **metastable**, and can be pushed to a number of alternative (meta)stable states that depend on the nature of the ecosystem itself.

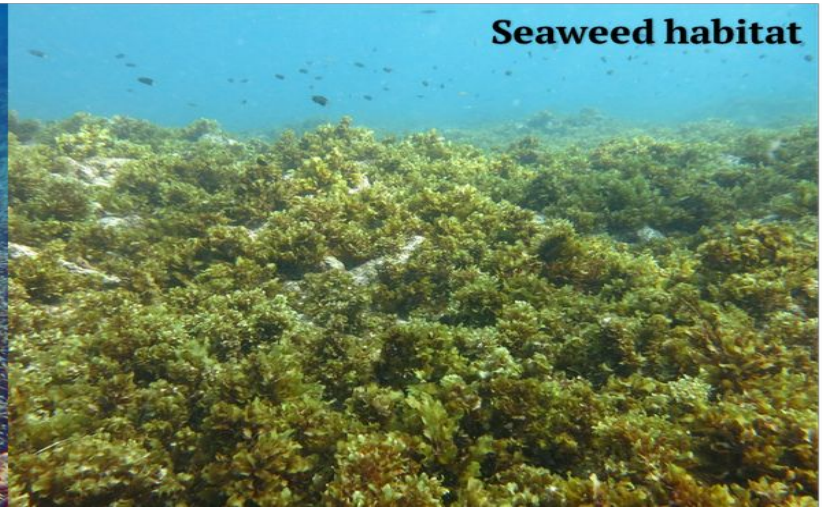
These situations can be represented by a ball moving between multiple valleys, the center of which represents a single **alternative stable state**, separated each other from the hills, the top of which is called tipping point. If the disturbance is strong enough to allow the ball to reach the tipping point, it can move to an alternative stable state. The higher are resilience and resistance of an ecosystem, the more difficult is the transition. Vice versa, the lower are resilience and resistance, the more vulnerable is an ecosystem.



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BEF: Ecosystem Disturbance And Recovery



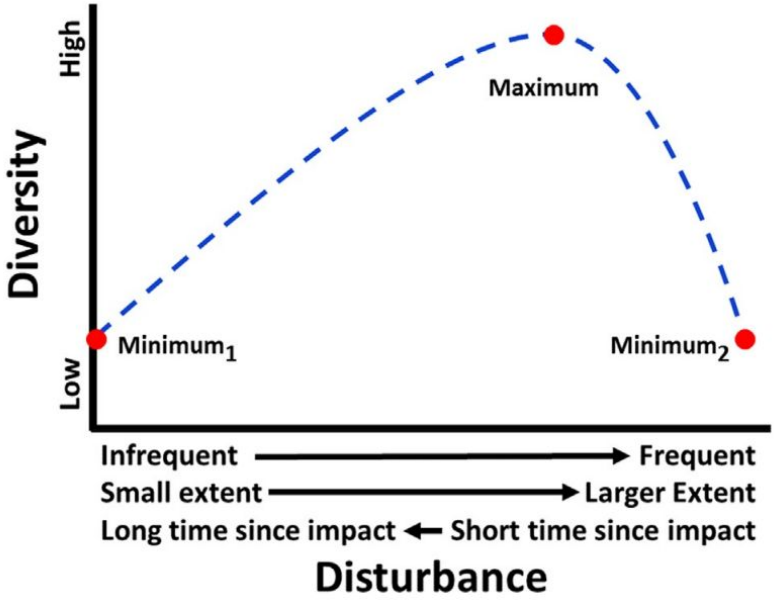


Coral reefs represent a clear example of resilience of an ecosystem. It can be a **high-resilience habitat**, when after a physical damage due to hurricanes, overfishing, careless tourism, corals can recover re-building the reef, or a **low-resilience habitat**, when, bleaching induces corals death and the seaweed takes over the reef. In this latter, coral reef and seaweed represent two alternative stable states of an ecosystem.

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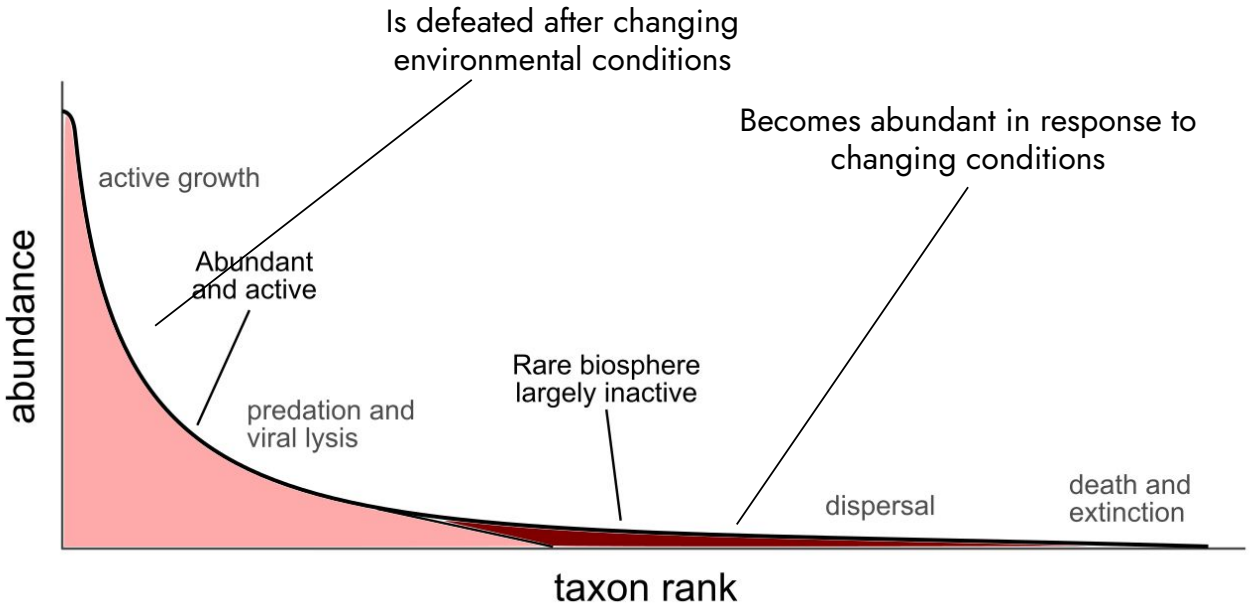
BEF: Intermediate Disturbance Hypothesis (IDH)

The intermediate disturbance hypothesis (IDH) suggests that species richness (and diversity) at the local scale reach its maximum values at intermediate values of disturbance frequency, intermediate times after a disturbance, and at intermediate spatial extents of disturbance. The idea is that recurrent disturbances create non-equilibrium conditions that reduce the probability of competitive exclusion between co-occurring species and consequently promote high species richness and diversity.
If the extent of disturbance is too low, the system becomes so stable over time that competition is too strong and only dominants species are able to survive. If the disturbance is too strong, only the most resistant and adaptable species can survive.
A perfect example of this situation is viral infection and "kill the winner" hypothesis.



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Role of rare and dormant species as a **reservoir** of functions and adaptations in response to environmental disturbance

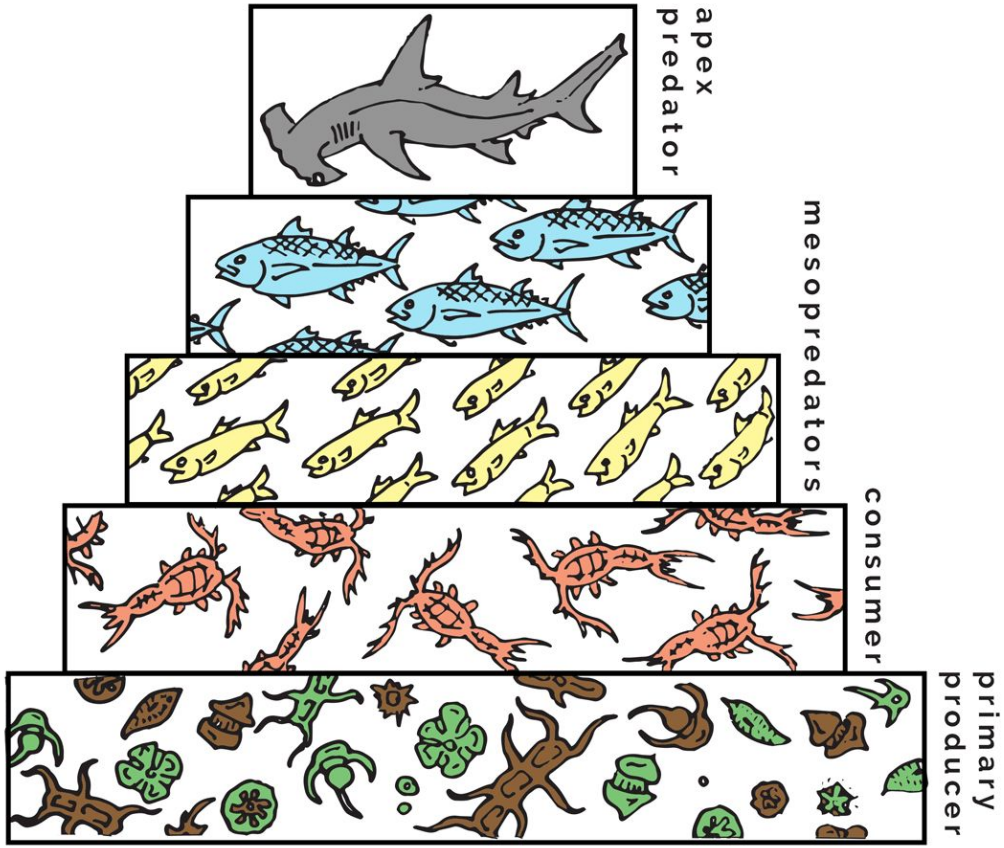


The role of the rare species is also that of **re-building the ecosystem** after a disturbance. The presence of rare species could indicate the **low resilience** of the ecosystem.

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Large Scale Marine Community Interactions

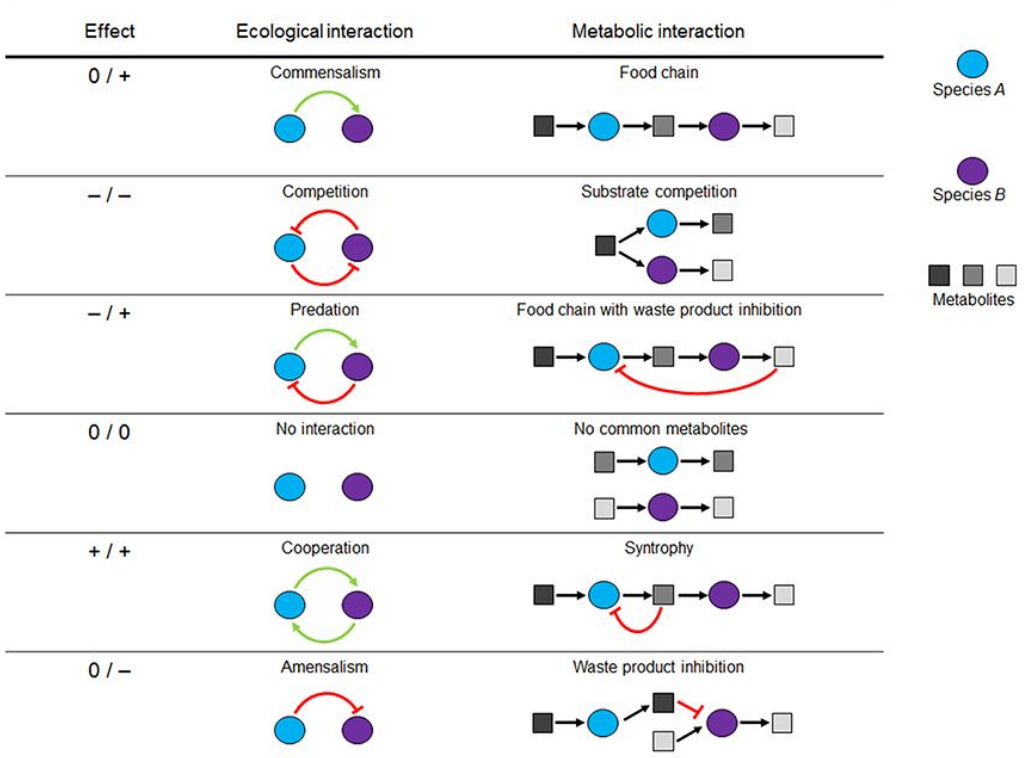
Marine food chain is commonly divided into trophic levels where organisms are grouped by the role they play, from bottom prays to top predators, and can be illustrated as a 4-level pyramid. First trophic level includes **primary producers**, such as phytoplankton, seagrass and protists. Second level represent the **primary consumers**, including herbivores and carnivores zooplankton and crustaceans. Then follow the intermediate or meso-**predators** and top or apex-predators, which close the food chain.



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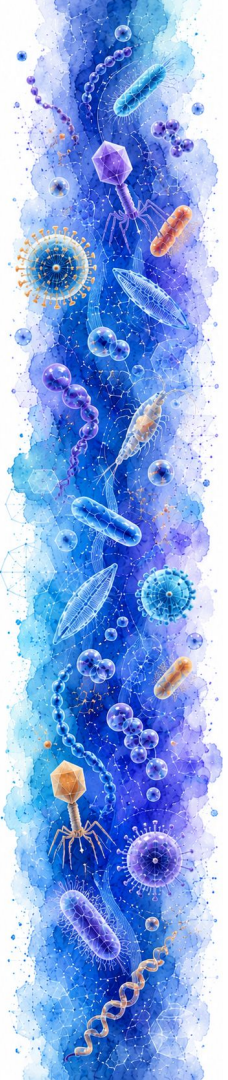
Small Scale Microbial Community Interactions

Species interactions in microbial communities can be either metabolism-based or be driven by ecological traits.
The net effect of the interactions between species can be positive, negative, or neutral resulting in many possible combinations of win, lose, or neutral outcomes.
Based on these interactions, the creation of both theoretical mechanistic models and experimental manipulation practices can contribute to our **understanding of microbial communities behavior** within and with their ecosystems.



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10.3389/fmicb.2016.00673



Community Interactions Shape Ecosystem Functioning

Trophic cascades are powerful indirect interactions that can control entire ecosystems, occurring when a trophic level in a food web is affected in terms of amount and diversity.

.**Bottom-up control** refers to the limitation of a prokaryotic community by the availability of energy or nutrients, which in turn affect higher trophic levels depending by primary producers. Changes in their population will affect the population of all the species in the upward food chain. An example is the sinking marine snow from a surface algal bloom floods seafloor sediments with energy-rich carbon, triggering a bottom-up population explosion of specialized heterotrophic bacteria (Bacteroidetes and Gammaproteobacteria).

.**Top-down control** refers to an increase of predators in high trophic levels inducing a limitation of a prokaryotic community. In this case, the abundance and diversity of primary producers at lower trophic levels depends on the predatory attitudes of primary consumers at higher trophic levels. Viral lysis and protist grazing are clear examples of top-down controls.

.**Sideways control** refers to a limitation of a prokaryotic community by the existent competition within the same trophic level. An example is the production by some microorganisms of antimicrobial compounds that inhibit or kill sensitive strains from their own or closely related species, a phenomenon known as chemical warfare.

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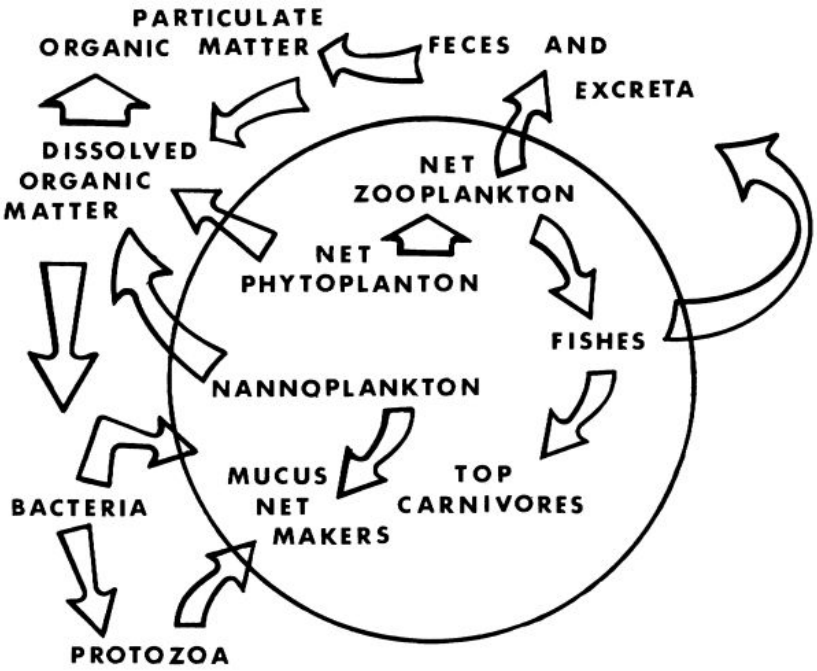
The Ocean's Food Web, A Changing Paradigm

Lawrence R. Pomeroy, 1974

"Basic to the understanding of any ecosystem is knowledge of its food web, through which energy and materials flow. Flux of both energy and essential elements shapes or limits ecosystems."

"Although the ocean's food web has been studied for more than a century, several recent discoveries lead us to believe that the classical textbook description of a chain from diatoms through copepods and krill to fishes and whales may in fact be only a small part of the flow of energy."

"Microorganisms (whose biomass is approximately equal to that of the net plankton) are greater movers of energy and materials because of their higher metabolic rate per unit mass."



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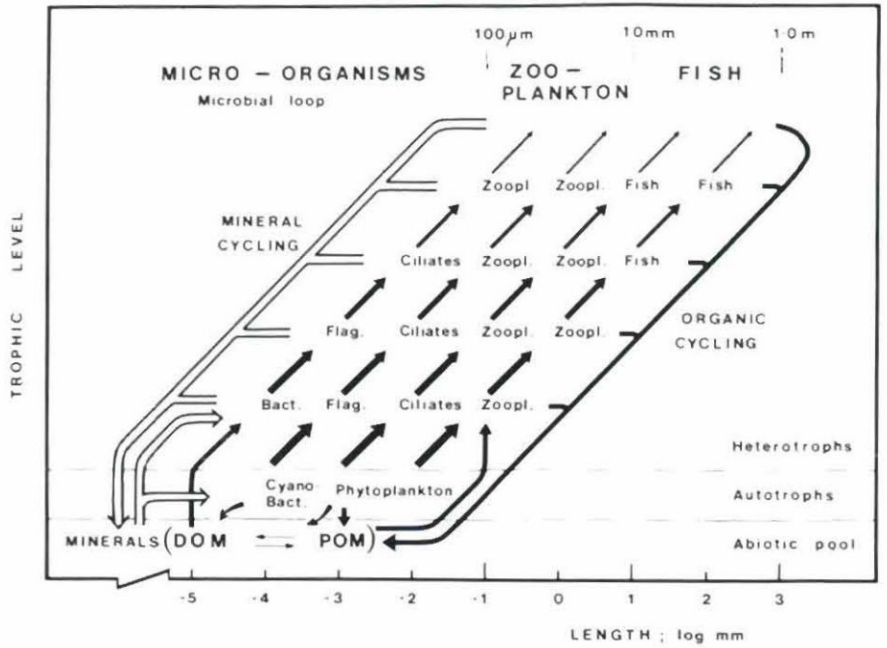
10.2307/1296885

The Ecological Role of Water-Column Microbes in the Sea*

Farooq Azam, 1983

"Bacterial biomass in the sea is related to phytoplankton concentration, since bacteria can utilise 10 to 50% of carbon fixed by photosynthesis. Evidence is presented to suggest that numbers of free bacteria are controlled by nanoplanktonic heterotrophic flagellates which are ubiquitous in the marine water column. The flagellates in turn are preyed upon by microzooplankton.

The regeneration of nutrients resulting from predation will provide a feedback of some of the material and energy flows within the microbial loop to the conventional planktonic food chain."



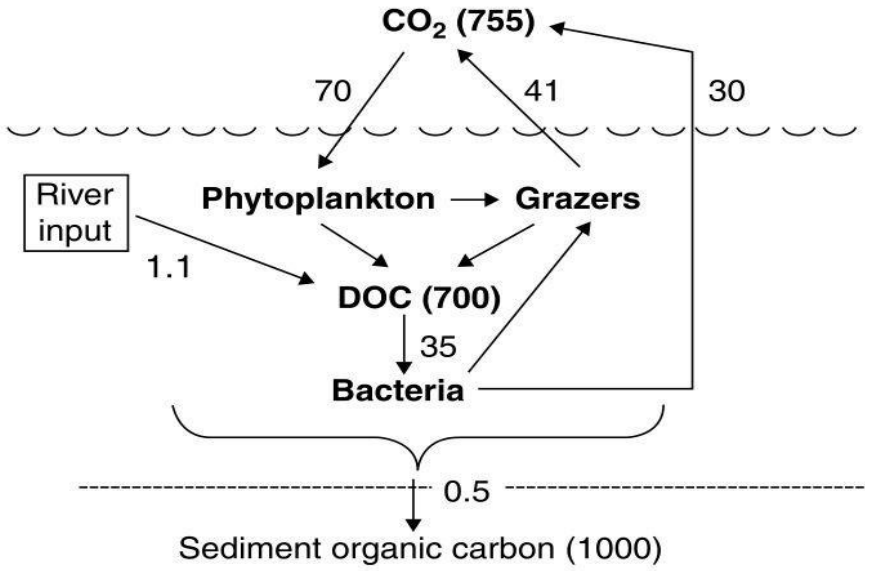
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10.3354/meps010257

The Microbial Loop

The microbial loop describes a trophic pathway in the marine microbial food web where, dissolved organic material (**DOM**) or carbon (**DOC**) **is returned to higher trophic levels via its incorporation into bacterial biomass, and then coupled with the classic food chain formed by phytoplankton-zooplankton-nekton.**

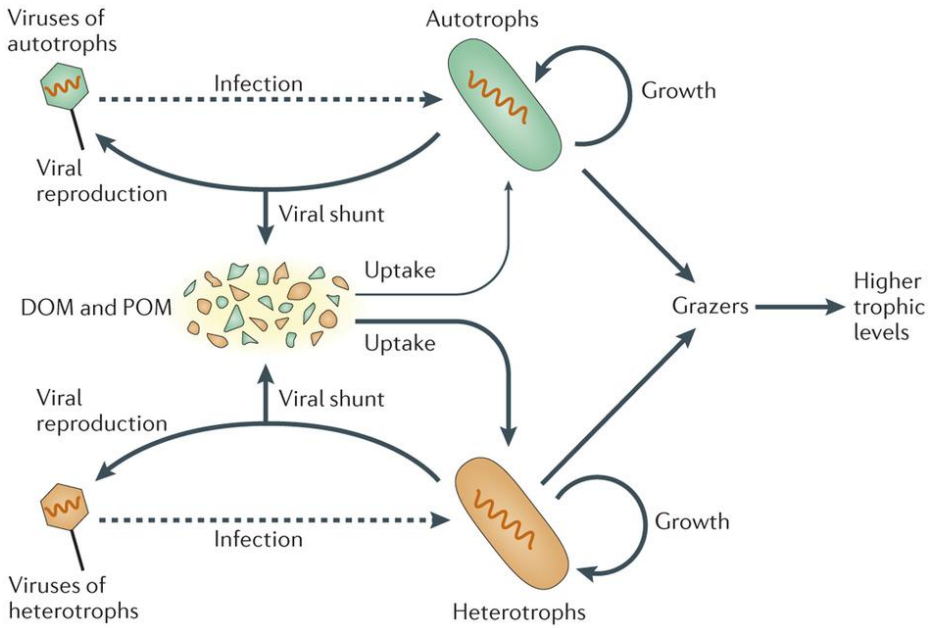
In general, DOC is introduced into the ocean environment from carbon fixation, cell lysis induced by grazing and viruses, exudation and excretion of organics by phytoplankton, the release of waste products by aquatic animals, or the breakdown or dissolution of organic particles from terrestrial sources. Heterotrophs synthesize lytic enzyme to decompose this particulate detritus and utilize this energy-rich matter for growth, reintroducing organic carbon into the food chain and thus resulting in additional energy becoming available to higher trophic levels.



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Viral Impact On Microbial Communities: Viral Shunt

When the microbial loop was first conceived, little was known regarding abundance and activity of marine viruses. Now, it is clear that viral lysis of primary producers and heterotrophic organisms act as a short circuit affecting the flow of nutrients toward higher trophic levels. In fact, by infecting and lysing prokaryotic cells, **virus are responsible for the effective generation of DOC that is available for microbial re-consumption**, thus recycling huge amounts of organic carbon (~0.37-0.63 Gt of carbon per year) and other nutrients in marine ecosystem. This process of virus-mediated recycling of organic matter is known as **viral shunt**.

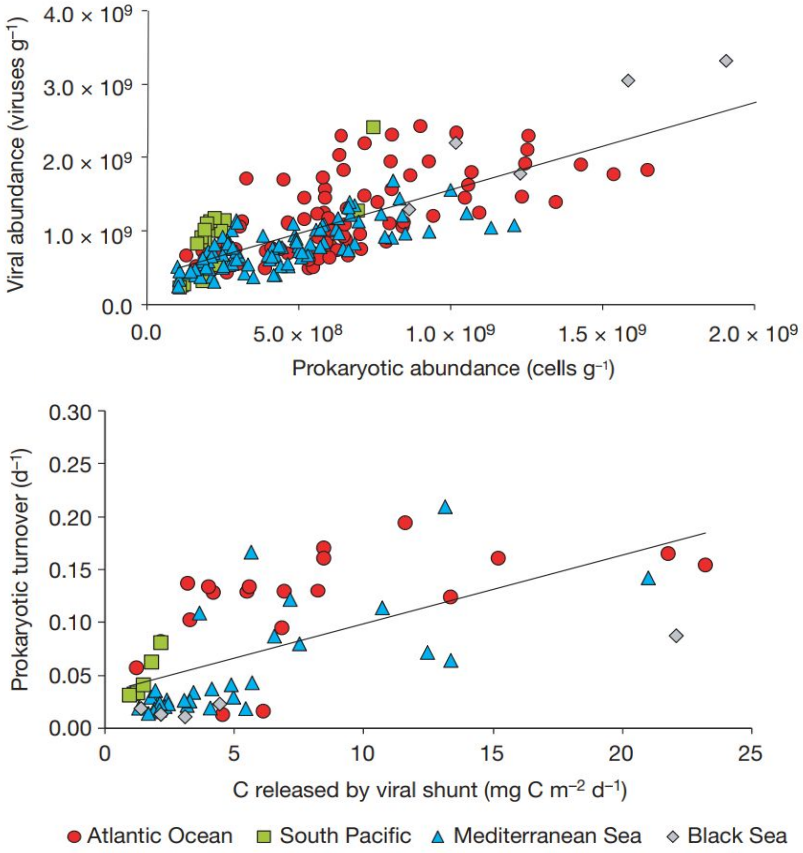


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Viral Impact On Microbial Communities: Viral Shunt

Deep-sea ecosystems cover about 65% of the Earth's surface and play an important part in biomass production and biogeochemical cycles on a global scale. Viral activity in deep-sea benthic ecosystems worldwide is extremely high, and the viral shunt is an essential source of labile organic detritus in the deep-sea ecosystems.

This process sustains a high prokaryotic biomass and provides an important contribution to prokaryotic metabolism, allowing the system to cope with the severe organic resource limitation of deep-sea ecosystems. Viruses have an important role in global biogeochemical cycles, in deep-sea metabolism and the overall functioning of the largest ecosystem of our biosphere.

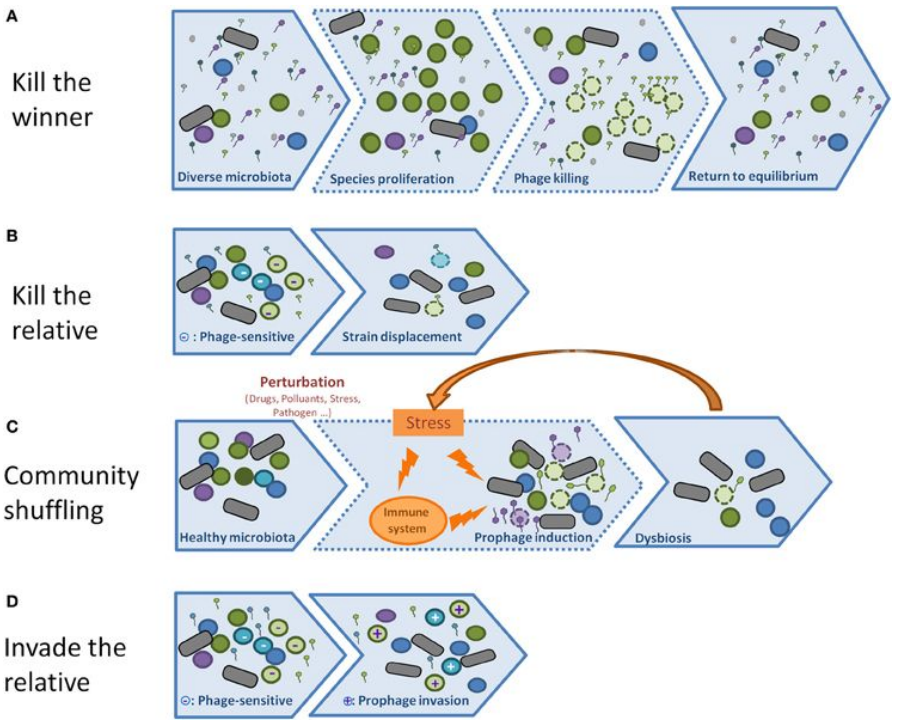


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10.1038/nature07268

Viral Impact On Microbial Communities

Viral lysis is one of the **top-down control** of prokaryotic and phytoplankton populations in marine environment, playing a fundamental role in shaping the community structure. Viruses can impact prokaryotic community in different alternative ways. Lytic viruses generally act by direct predation, infecting and multiplying in susceptible hosts. Temperate viruses can also lyse their previously lysogenic host upon induction. Alternatively, temperate viruses could also impact the ecosystem without killing their hosts, but by carrying genes within them, they can modify host phenotypes.



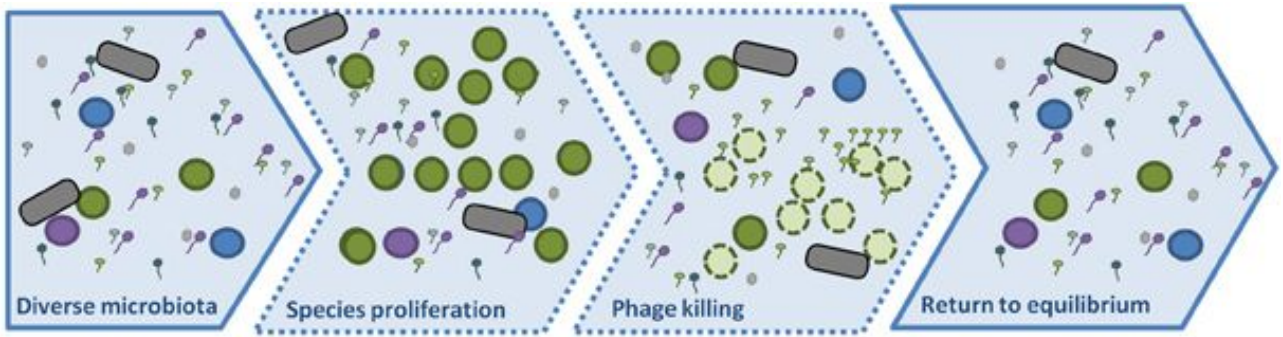
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10.3389/fcimb.2014.00039

Viral Impact On Microbial Communities: Kill The Winner Hypothesis

A

Kill the winner

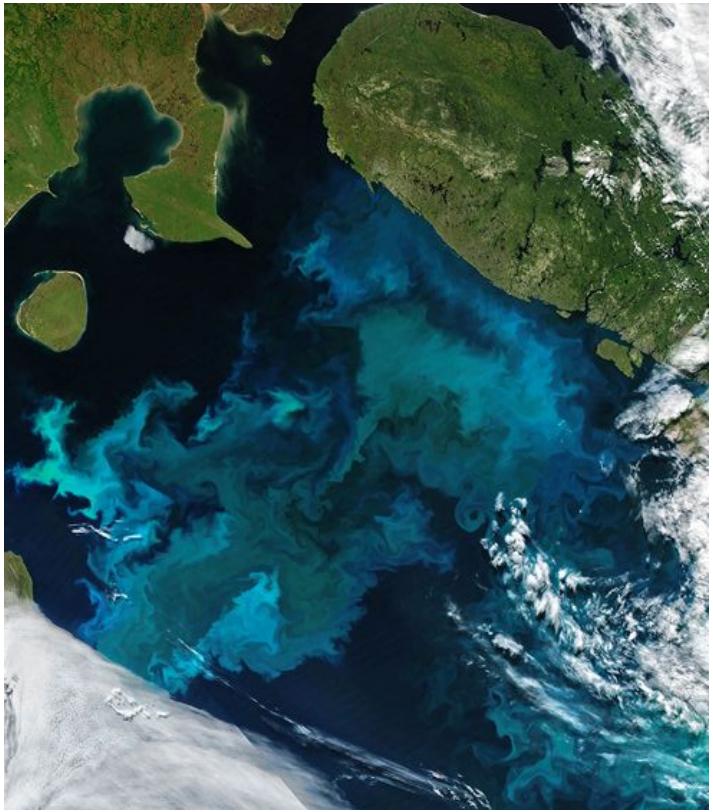


The "**kill the winner**" hypothesis is an ecological model of population growth developed by Tron Frede Thingstad in 2000 based on a mathematical model of the traditional host-parasite dynamics involving prokaryotes, viruses and protozoans. Phages cannot actively move toward their bacterial preys, they rely entirely on chance to encounter their hosts through random collision, and their ability to infect and reproduce is determined by host density. In this scenario, the most successfully competitive hosts (the "**winners**" with greater growth rate and abundance) will have the highest contact with viruses that infect them, resulting in greater infection rates and thus greater mortality that will reduce the abundance of a dominant host and allow other hosts to increase in prevalence. Therefore, in the "kill the winner" scheme, viruses only start to infect when their hosts reach the reproduction threshold, above which phages can absorb and predate them, they will lyse, and the system will shift back to its initial state.

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Viral Impact On Microbial Communities: Kill The Winner Hypothesis

The "kill the winner" hypothesis suggests that for a population of phages to multiply, the bacterial population must be above a critical density termed replication threshold. As a consequence, phage multiplication is possible only if the population of bacterial prey is high. A clear example of "kill the winner" hypothesis is the viral-driven bloom collapse of *Emiliania huxleyi*. Massive natural blooms of the coccolithophore develop annually in many temperate and sub-temperate coastal and oceanic waters. At high host densities or bloom climax Coccolithoviruses (EhVs) initiate lysis rapidly after establishing infection. *E. huxleyi* bloom collapse generally leads to the release of millions of coccoliths (calcium carbonate shells). This event attributes a pivotal role to *E. huxleyi* in the global carbon cycle, being the largest global producer of calcium carbonate and, hence, a major sink for CO_2.



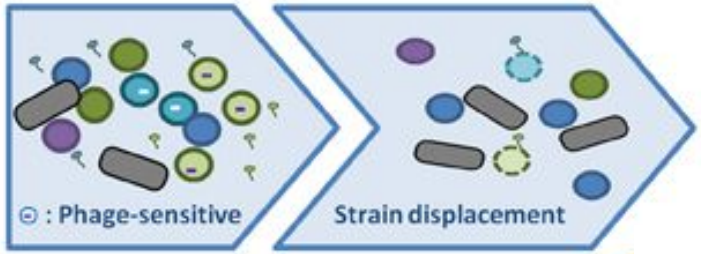
Emiliania huxleyi in Barents Sea, Russia

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Viral Impact On Microbial Communities: Kill The Relative Hypothesis

B

Kill the relative



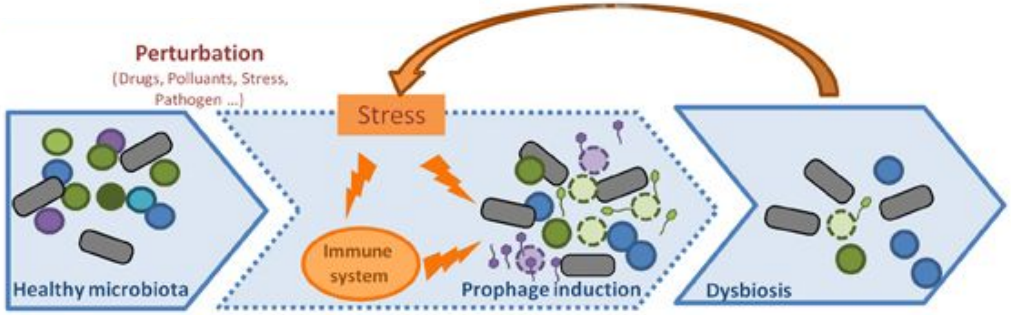
In some cases lysogenic bacteria can use their prophages as biological weapons against their main competitors, related bacteria that use the same ecological niche. Rare, spontaneous prophage induction in a small fraction of the lysogenic population can trigger an outbreak among susceptible bacterial competitors, which become factories producing more phage. As the prophage provides immunity to its carrier bacteria against further infection by this phage, the lysogenic bacteria are protected. Thus, by killing competitor bacteria, prophages can indirectly benefit their bacterial host. Therefore, In the "kill the relative" scheme, phages do not need to be abundant, they are released by lysis of some lysogen strains and kill their relatives that are not resistant to the phage.

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Viral Impact On Microbial Communities: Community Shuffling Model

c

Community shuffling



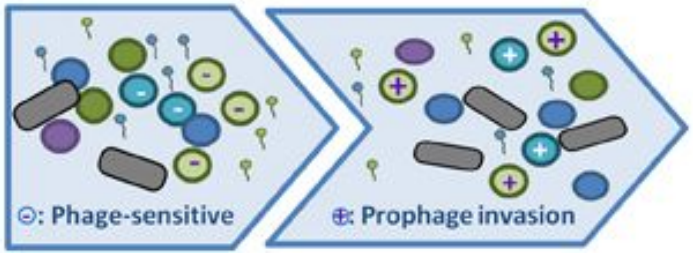
Rather than being beneficial to their bacterial host, prophages can also be detrimental as their induction results in host lysis. A variety of environmental factors can change induction from a rare and stochastic event to a deterministic process. In oceans, prophages have been compared to molecular time bombs that can be triggered by changes in salinity or by various pollutants. In the "community shuffling" scenario, temperate phages act negatively on their host by killing them upon sensing a mild stress that would not have killed a non-lysogen. A positive feedback loop may even take place if the massive lysis leads to an host reaction, such as inflammation (orange arrow). The net result is a global displacement of population, and possibly dysbiosis.

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Viral Impact On Microbial Communities: Invade The Relative Model

D

Invade the relative



An important alternative to the above mentioned "kill the relative" model is therefore an "invade the relative" situation, whereby prophages behave as rampant invaders, and thereby contribute to the emergence of new bacterial strains. Such horizontal transfer can induce important changes in bacterial phenotypes, often related to a shift toward pathogenicity, such as increased virulence due to toxin production, or improved colonization abilities, due to resistance to oxidative stress, bile salts, or biofilm formation. Therefore, in the "invade the relative" scheme temperate phages propagate by infecting new hosts without lysing them, but by establishing lysogeny, actively contributing to the generation of new gene - and genome - combinations within the microbial community.

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Viral Impact On Microbial Communities: Red Queen Hypothesis

Phages replicate more rapidly than their hosts, and lytic infection results in death of the host, why don't phages kill all the hosts and cause their own extinction in the process?
A common explanation is that there is a **constant co-evolution of the host and virus**, an "arms race" via mutation and counter-mutation, resulting in continual selection pressure for the host to evolve resistance and for the virus to evolve to overcome that resistance.



"Well, in our country," said Alice, still panting a little, "you'd generally get to somewhere else — if you run very fast for a long time, as we've been doing."

"A slow sort of country!" said the Queen. "Now, here, you see, it takes all the running you can do, to keep in the same place. If you want to get somewhere else, you must run at least twice as fast as that!"

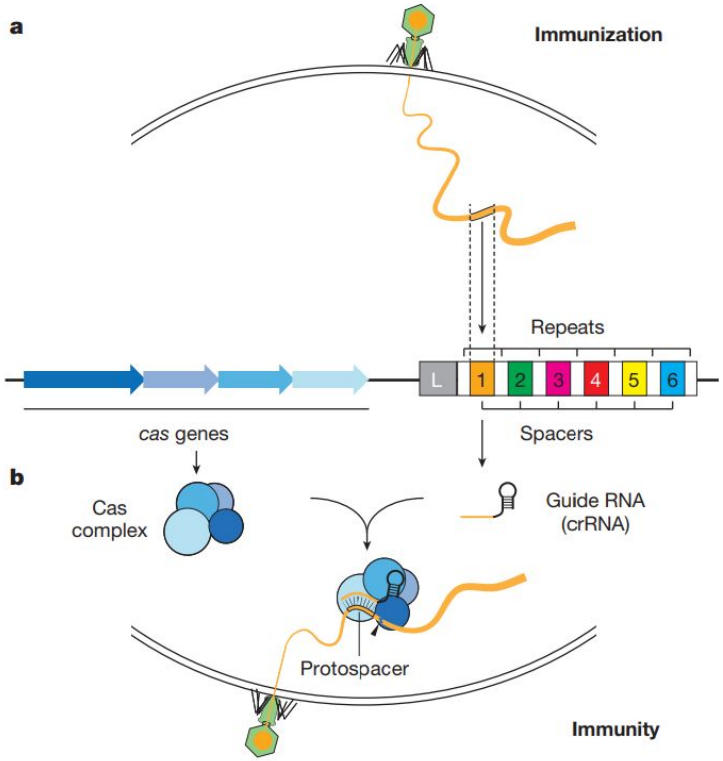
— Lewis Carroll, *Alice through the Looking Glass*

The Red Queen hypothesis is an evolutionary hypothesis which proposes that organisms must constantly adapt, evolve, and proliferate not merely to gain reproductive advantage, but also simply to survive while pitted against ever-evolving opposing organisms in an ever-changing environment.

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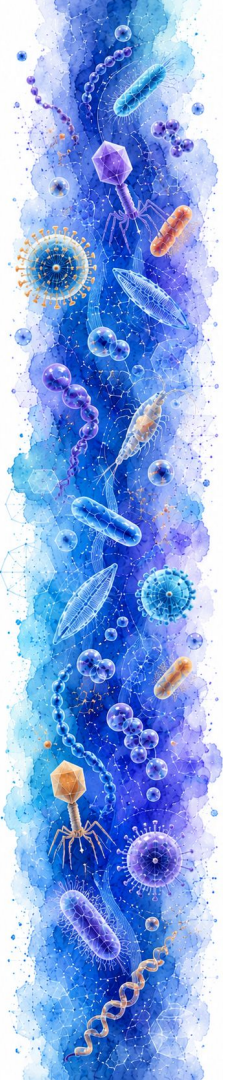
Mechanisms Of Antiviral Defence In Prokaryotes

To combat hostile viruses, bacteria and archaea have evolved a unique antiviral defense system composed of clustered regularly interspaced short palindromic repeats (**CRISPRs**), together with a CRISPR-associated genes (**Cas**). The **CRISPR/Cas9 system** develops an adaptive immune resistance to foreign plasmids and viruses by creating site-specific DNA double-stranded breaks (DSBs). Upon viral injection, a small sequence of the viral genome, known as a **spacer**, is integrated into the CRISPR locus to immunize the host cell. Spacers are translated into small RNA guides that direct the cleavage of the viral DNA by Cas nucleases. Immunization through spacer acquisition enables a unique form of evolution whereby a population not only rapidly acquire resistance to its predators but also passes this resistance mechanism vertically to its progeny.



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10.1038/nature15386



This Week Read

Azam, F., Malfatti, F. Microbial structuring of marine ecosystems. Nat Rev Microbiol 5, 782–791 (2007). https://doi.org/10.1038/nrmicro1747